

NE 470/570 Monte Carlo Methods for Radiation Transport Fall 2026 Semester

Homework 3: Statistical moments, limit theorems, confidence level, normality testing
Deadline: 10/01/2026 11:59 PM

Problem 1

Random variable X has a density function expressed by $f(x) = 1 + x$, $0 \leq x \leq 1$. If function $g(X)$ is expressed by $g(X) = X + cX^2$, determine the parameter c such that the variance of $g(X)$ is minimized.

Problem 2

Write a program to sample the path length that a particle travels in a medium with $\Sigma = 1.0 \text{ cm}^{-1}$. Considering that the path length is sampled from a PDF given by $p(r) = \Sigma \cdot e^{-\Sigma r}$, estimate the sample mean and sample variance for the following combinations of trials and histories:

- Total number of experiments, i.e., “trial” $\times$ “history per trial”, is fixed to 10,000. Consider the following history/trial ratios: 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000;
- Number of “history per trial” is fixed to 10. Consider the following number of “trials”: 10, 20, 30, 40, 50, 100, 200, 500 and 1000;
- Number of “trials” is fixed to 20. Consider the following “history per trial” ratios: 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000 and 5000.

You should use a proper PRNG to generate MC samples. Tabulate the results like Table 4-4 in the slides, and plot the sample variance w.r.t. the number of histories per trial. Discuss your observations, especially regarding the Central Limit Theorem.

Problem 3

Use a proper PRNG to generate samples that follow a standard uniform distribution. You will need a relatively large number of samples, such as 2^{20} . Define N as the number of samples in each trial, T as the number of trials. In each trial, randomly select N samples from the large sample set WITHOUT replacement, and calculate the mean value. You will obtain T different values of the sample mean. For $N = 100, T = 100$ and $N = 20, T = 500$, perform the following analysis:

- Compare the normalized histogram of these T samples with the corresponding PDF of the Gaussian distribution.
- Normality test with skewness coefficient. Compare it with the two threshold values. Calculate with `scipy.stats.skew`, as well as the formula provided in the lecture slides. Does `scipy.stats.skew` use the biased sample variance or unbiased sample variance?
- Normality test with Shapiro-Wilk test. What is your p-value? Explain your test results. You can only use the `stats.shapiro` function, there is no need to manually calculate the test statistic.

Problem 4

Using the t-table shown below, evaluate the confidence intervals:

1. The targeted length of screws produced by a certain factory is 5 cm. After drawing a sample of 30 screws, the mean length $\bar{x}$ is 4.8 cm, the unbiased standard deviation of these samples S_x is 0.4 cm. What are the 95% and 90% confidence intervals for the population mean?
2. If the true standard deviation σ_x is known to be 0.5 cm, and $n = 50$ samples are used, given $\bar{x}$ as 4.8 cm, what are the 95% and 90% confidence intervals for the population mean?

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										